

HARBIN, China

WMO: 509530

Lat: 45.9340N

Lon: 126.5810E

Elev: 118

StdP: 99.92

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.4	(c) -25.2	(d) -30.9	(e) 0.2	(f) -26.3	(g) -28.9	(h) 0.3	(i) -24.2	(j) 7.2	(k) -9.8	(l) 6.3	(m) -10.7	(n) 1.4	(o) 180	(p) 0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.8	(c) 31.5	(d) 20.6	(e) 30.1	(f) 21.1	(g) 28.8	(h) 20.9	(i) 24.4	(j) 28.3	(k) 23.6	(l) 27.3	(m) 22.8	(n) 26.5	(o) 3.6	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	18.3	26.6	22.4	17.3	25.9	21.6	16.5	25.3	74.9	28.4	71.5	27.5	68.4	26.5	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 6.5	(c) 5.8	(e) -29.6	(f) 34.5	(g) 3.2	(h) 1.8	(i) -31.9	(j) 35.8	(k) -33.7	(l) 36.8	(m) -35.5	(n) 37.8	(o) -37.9	(p) 39.1		
(a) 7.5	(b) 6.5	(c) 5.8	(e) -29.7	(f) 25.5	(g) 3.1	(h) 1.2	(i) -32.0	(j) 26.4	(k) -33.8	(l) 27.0	(m) -35.6	(n) 27.7	(o) -37.9	(p) 28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-17.1	-11.9	-2.7	7.7	15.4	21.5	23.7	21.9	15.8	6.9	-4.7	-14.5		
	DBStd	15.00	4.53	5.52	5.84	4.94	4.21	3.40	2.29	2.60	3.59	4.76	6.20	4.95		
	HDD10.0	3272	841	612	393	101	4	0	0	0	2	119	441	758		
	HDD18.3	5192	1099	846	652	319	111	10	1	4	91	353	691	1016		
	CDD10.0	1547	0	0	1	33	173	345	425	369	177	24	1	0		
	CDD18.3	424	0	0	0	1	21	105	167	115	15	0	0	0		
	CDH23.3	3293	0	0	0	18	243	911	1269	741	109	2	0	0		
	CDH26.7	903	0	0	0	2	69	312	356	153	11	0	0	0		
(14) Wind	WSAvg	2.6	2.2	2.5	3.0	3.4	3.2	2.7	2.4	2.3	2.3	2.6	2.7	2.3		
(15) Precipitation	PrecAvg	527	4	5	12	21	50	97	139	105	47	26	14	8		
	PrecMax	823	13	18	38	62	113	209	357	238	86	55	46	20		
	PrecMin	384	0	0	1	1	10	16	45	34	4	9	1	1		
	PrecStd	99	3	5	9	15	29	46	59	51	23	12	10	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.6	5.2	16.0	25.7	31.5	34.9	33.2	31.1	28.2	22.8	13.4	3.5		
		MCWB	-3.9	1.2	7.5	12.2	16.1	19.2	22.5	23.7	20.1	14.7	8.5	1.8		
	2%	DB	-5.3	2.4	11.6	22.2	28.6	31.9	31.3	29.7	26.1	19.7	10.0	-1.7		
		MCWB	-7.0	-1.0	5.0	10.9	15.5	18.9	22.7	22.8	18.6	12.8	5.8	-3.4		
	5%	DB	-7.7	-0.2	8.6	19.6	26.1	30.2	29.9	28.4	24.4	17.0	7.2	-4.5		
		MCWB	-9.1	-2.9	2.9	9.7	14.5	19.1	22.6	22.1	17.4	10.9	3.7	-6.0		
	10%	DB	-10.0	-3.0	6.1	16.9	23.5	28.2	28.7	27.2	22.7	14.7	4.1	-6.8		
		MCWB	-11.1	-5.3	1.4	8.4	13.5	18.5	22.4	21.3	16.2	9.6	1.3	-8.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.7	1.5	7.9	13.7	18.6	23.0	26.0	25.4	22.0	15.6	9.2	1.6		
		MCDB	-1.9	4.5	14.6	22.5	26.9	28.6	29.3	29.3	25.7	21.0	12.6	3.3		
	2%	WB	-7.0	-0.6	5.3	11.9	16.9	21.9	24.8	24.2	20.2	13.6	6.1	-3.3		
		MCDB	-5.5	2.2	10.7	20.2	24.3	27.5	28.6	28.0	24.3	18.2	9.2	-2.0		
	5%	WB	-9.1	-3.0	3.4	10.2	15.6	20.8	24.0	23.3	18.8	11.9	3.7	-5.9		
		MCDB	-7.8	-0.2	8.0	18.3	23.3	26.2	27.8	26.8	22.7	15.9	7.0	-4.5		
	10%	WB	-11.1	-5.2	1.7	8.7	14.4	20.0	23.3	22.5	17.5	10.1	1.4	-8.1		
		MCDB	-10.1	-3.1	5.8	16.0	22.0	25.4	27.1	25.7	21.2	14.2	3.9	-6.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.6	9.9	10.8	10.6	9.5	7.8	8.3	10.7	10.0	8.2	8.4		
		MCDBR	11.5	11.7	13.0	14.6	14.9	12.4	9.8	9.9	12.6	13.1	11.3	10.4		
	5% WB	MCWBR	10.1	9.4	8.0	7.6	6.1	4.4	3.7	4.2	5.9	7.3	8.2	9.3		
		MCWBR	11.5	11.7	12.3	13.0	11.7	9.1	7.8	7.9	10.3	11.5	10.9	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.342	0.416	0.518	0.580	0.620	0.584	0.514	0.461	0.455	0.373	0.335		
	taud		2.077	2.057	1.941	1.780	1.707	1.721	1.835	1.991	2.037	1.935	2.058	2.064		
	Ebn at Noon		732	806	791	741	711	685	705	741	742	662	662	666		
	Edh at Noon		98	122	160	206	229	227	201	165	144	138	100	90		
(44) All-Sky Solar Radiation	RadAvg		2.09	3.23	4.32	4.90	5.47	5.63	5.16	4.81	4.23	2.95	1.94	1.65		
	RadStd		0.20	0.22	0.33	0.36	0.51	0.59	0.50	0.35	0.33	0.20	0.15	0.10		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page